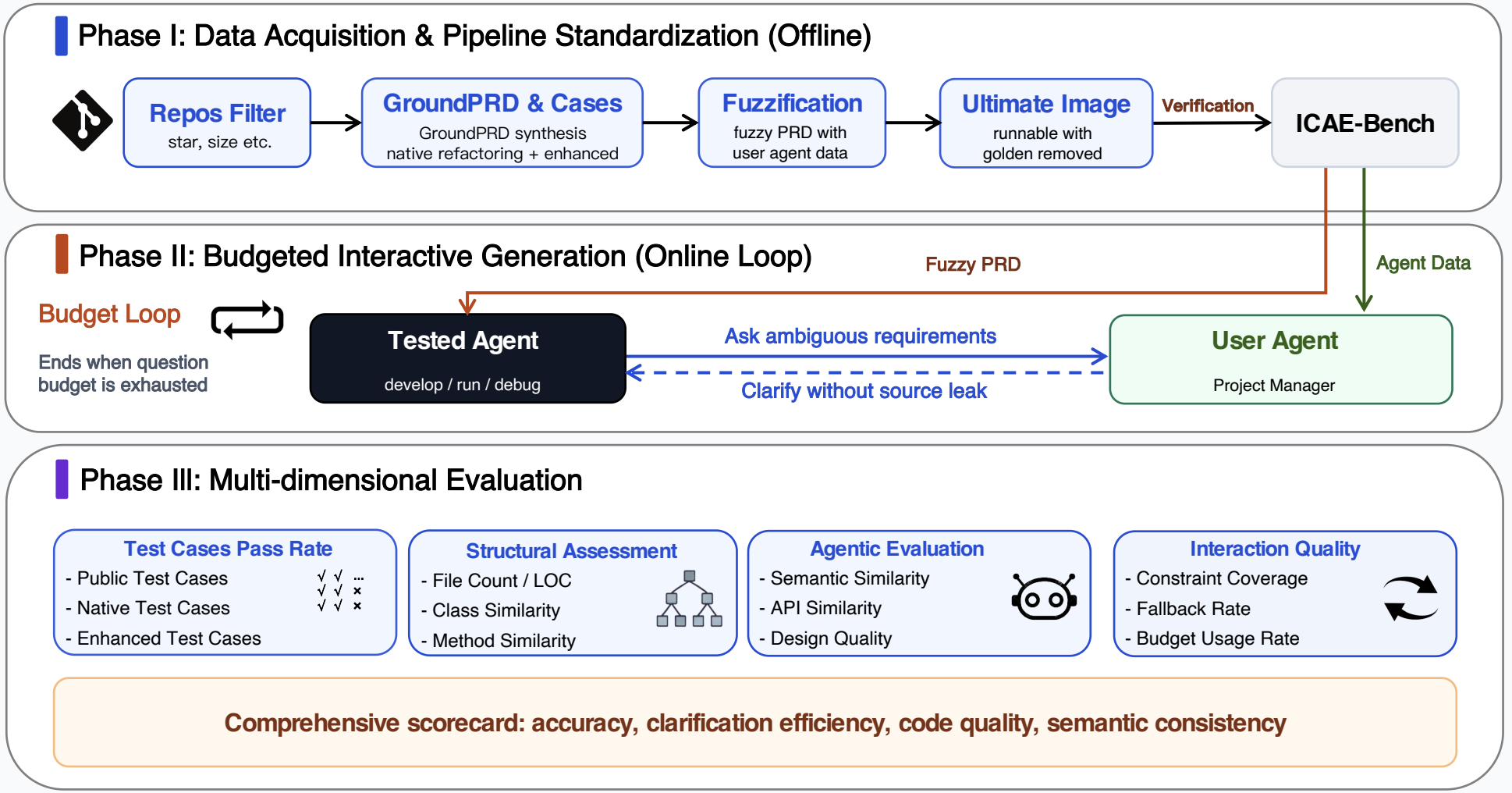

Phase I: Data Acquisition & Pipeline Standardization (Offline)
Repos Filter
star, size etc.
GroundPRD & Cases
GroundPRD synthesis
native refactoring + enhanced
Fuzzification
fuzzy PRD with
user agent data
Ultimate Image
runnable with
golden removed
Verification
ICAE-Bench
Phase II: Budgeted Interactive Generation (Online Loop)
Fuzzy PRD
Agent Data
Budget Loop
Ends when question
budget is exhausted
Tested Agent
develop / run / debug
Ask ambiguous requirements
Clarify without source leak
User Agent
Project Manager
Phase III: Multi-dimensional Evaluation
Test Cases Pass Rate
- Public Test Cases
- Native Test Cases
- Enhanced Test Cases
Structural Assessment
- File Count / LOC
- Class Similarity
- Method Similarity
Agentic Evaluation
- Semantic Similarity
- API Similarity
- Design Quality
Interaction Quality
- Constraint Coverage
- Fallback Rate
- Budget Usage Rate
Comprehensive scorecard: accuracy, clarification efficiency, code quality, semantic consistency